SAEON Graduate Attributes and Portfolio Evidence

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The project applies computing and electronics to the need for accessible environmental observations. This document explains how the four graduate attributes relate to the work and identifies the evidence needed for the portfolio. Each member should adapt the reflection to their own contribution and attach genuine examples.

The ESP32 WROOM-32 reads the TR-525M rain gauge. DHT11 is used for testing; SHT31 is the temperature and humidity sensor selected for the final product. Firebase is the database platform.

Ubuntu

The project begins with the needs of SAEON researchers who need access to observations from remote sites. A useful system must make the readings understandable and accessible, rather than simply collecting data. Clear units, current timestamps, mobile access and honest indications of missing data help people use the information responsibly.

The team's work depends on cooperation between hardware, backend, frontend, research and coordination roles. Interpersonal skills are needed when explaining a fault, asking for help or agreeing priorities. Interprofessional skills are needed when translating between technical implementation and the requirements of environmental researchers. We should check user needs directly and respect different knowledge, working practices and cultural perspectives rather than assume them.

Evidence to attach: a genuine requirements discussion or feedback record, an example of teammate collaboration and an example of engagement with a supervisor or environmental stakeholder. The charter establishes responsibilities but does not, on its own, prove these interactions.

Knowing Technological Capability and Foresight

The planned system connects ESP32 sensor readings, cellular communication, MQTT, Firebase and a dashboard. It shows how knowledge from programming, data handling and electronics can be combined to address a practical monitoring problem. DHT11 supports initial testing, while SHT31 is the selected sensor for the final product. Recording that distinction helps avoid presenting prototype results as proof of final-product performance.

Remote monitoring could reduce the need to retrieve observations manually and make environmental data available sooner. That is the intended benefit; it must be evaluated through actual deployment and user feedback. Calibration, access control, power consumption and maintainability are part of planning for long-term use.

Evidence to attach: architecture and wiring diagrams, DHT11 test output, SHT31 verification when available, a Firebase record matched to a displayed reading, and an explanation of the intended environmental benefit.

Resilience and Problem Solving Capability

The system requires several connected parts to work together. A missing dashboard reading could arise from the sensor, power supply, cellular network, MQTT delivery, ingestion service, Firebase access rules or frontend. A useful problem-solving approach is to test each stage and compare timestamps and station identifiers before changing several components at once.

Ethan's 4–6 August entries report unavailable sensors, a breadboard capacity problem and an HTTP connection problem. The next steps in those entries include improving the physical setup and moving to Firebase. These entries provide a starting point for discussing obstacles and perseverance, but they do not by themselves establish that every problem was resolved. The reflection should identify the actual action, evidence and resulting lesson.

The sprint breakdown and named leads support organisation. Financial capability should be demonstrated with a real component budget, purchase record or cost comparison. No verified budget or savings figure is asserted here. Innovation lies in adapting an integrated monitoring approach to the project need; research evidence should explain the alternatives considered and why the chosen configuration was suitable.

Evidence to attach: dated troubleshooting notes, before-and-after results where available, task reprioritisation, a real bill of materials or budget, and component or communication research. Explain which outcome occurred and what remains unresolved.

Ethical Capability and Respect

Environmental data should be presented accurately, including uncertainty and missing observations. The project therefore needs to identify the sensor used, retain measurement units and timestamps, distinguish prototype testing from final validation and keep incomplete field work visible. Reporting an unverified result as successful would weaken confidence in the evidence.

Firebase access should follow the agreed permissions, and privileged credentials should not be published in the portfolio. Team contributions and external sources should be acknowledged. These principles support ethical decisions about data quality, access and honest reporting; the portfolio needs an actual example showing how a member applied them.

The guide also asks for an example of acting on unethical practices. No specific incident is established by these documents. Each student should describe a genuine relevant incident and their response if one occurred. Otherwise, state that none is recorded and ask the lecturer what alternative evidence is acceptable. Do not create an incident to fill this requirement.

Evidence to attach: actual access-control tests, an example of a corrected reporting claim or attribution, and a genuine ethical decision or incident record where available.

Individual contributions and reflection prompts

Use the records below as a starting point for an individual reflection. Describe the situation, your responsibility, the action you actually took, the result and what you would improve. The role allocation is documented; a listed activity is not proof that the individual completed it.

Member and role	Focus for a personal example	Evidence to link
Redah Gamieldien Project Leader	Coordinating hardware work, separating testing from final sensor acceptance, and managing open deployment actions.	Confirmed log entries, wiring, sensor checks and acceptance decisions.
Qaasim Isaacs Backend and testing	Investigating data-delivery problems, validating Firebase records and checking access permissions.	Confirmed log entries, ingestion code, Firebase records and test logs.
Angelo Adams Frontend and design	Making readings understandable across devices and showing stale or missing data clearly.	Confirmed log entries, interface screenshots, user feedback and responsive checks.
Ethan Williams Research and writing	Documenting obstacles, comparing component choices and keeping technical explanations consistent.	Dated logbook records, research notes, diagrams and document versions.
Rameez Karriem Agile lead and presenter	Coordinating contributions, making blockers visible and explaining limitations in the presentation.	Confirmed log entries, board history, review notes and presentation records.